



Lecture Activity 3

IT325 Machine Learning | Comparing Models and Reading Evaluation Metrics

Type	Lecture Activity
Topic Focus	Model comparison and evaluation for classification
Time Allotment	45 to 60 minutes
PIT Connection	Students learn how to justify model choice using evidence
Mode	Individual or pair discussion

Learning Outcomes

1. Explain why more than one model should be tested in a classification problem.
2. Differentiate accuracy, precision, recall, F1 score, and confusion matrix results.
3. Interpret model evaluation results and justify an initial best model.

Activity

Students analyze sample evaluation results from several classifiers and decide which model is most suitable for a given text classification task. The emphasis is on reasoning, not yet on coding.

Suggested Discussion Points

1. Why using only one model may lead to a weak PIT result

Using and relying on only one model provides no basis for comparison, in result this weakens the credibility of the output. Without testing multiple models, we cannot know if the model we have used is truly the best model for the dataset.

2. Why accuracy alone is not enough in many classification tasks

Accuracy measures the percentage of correct predictions but it is often misleading when classes are imbalance.

3. How false positives and false negatives affect decision making

The cost of each type of error depends on the context. In medical classification, false negatives are generally more dangerous because missing a real diagnosis can lead to serious harm.



UNIVERSITY OF SCIENCE AND TECHNOLOGY OF SOUTHERN PHILIPPINES

Alubijid | Balubal | Cagayan de Oro | Claveria | Jasaan | Oroquieta | Panaon | Villanueva

4. How evaluation metrics help in choosing models fairly

Evaluation metrics helps in choosing models fairly by providing an objective, standardized way to compare models. Instead of selecting a model based on intuition or a single number and metrics. This allows for a fair, evidence-based comparison across models so the final selection can be justified with data rather than preference.

In Class Analysis Task

1. Review a sample table of results from five models.

Model Evaluation Results (Positive class = Stroke)				
	Accuracy	Precision	Recall	F1-Score
Model				
Logistic Regression	0.4880	0.4849	0.4518	0.4678
Linear Regression	0.4880	0.4849	0.4518	0.4678
Decision Tree	0.4900	0.4889	0.5308	0.5090
Random Forest	0.4997	0.4975	0.4625	0.4794
K-Nearest Neighbors	0.4957	0.4937	0.4973	0.4955

2. Identify the model with the highest accuracy.

The model with the highest accuracy is **Random Forest** with **0.4997**.

3. Identify the model with the highest recall.

The model with the highest recall is **Decision Tree** with **0.5308**.

4. Identify the model with the most balanced performance.

The model with the most balanced performance is **Decision Tree** with **0.5090**.

5. Write a short justification for the model you would carry forward for tuning.

Since I am dealing with medical diagnosis tasks, **Recall** is objectively the most critical metric. **Decision Tree** have the highest recall with **0.5308** meaning that the model is better at identifying actual stroke cases and carries a lower risk of False Negatives. **Decision Tree** also have the most balanced performance with its **F1-Score** being at **0.5090**.

Reflection Prompt

A model with the highest accuracy is not always the best model. Explain this statement using a classification example related to your PIT dataset.



UNIVERSITY OF SCIENCE AND TECHNOLOGY OF SOUTHERN PHILIPPINES

Alubijid | Balubal | Cagayan de Oro | Claveria | Jasaan | Oroquieta | Panaon | Villanueva

In my dataset and with the table presented above, **Random Forest** holds the title of highest accuracy. However, because this dataset concerned with the goal of life saving, **Recall** matters more than accuracy. While Random Forest was most accurate, the Decision Tree achieved the highest Recall and the best F1-Score. In a medical classification tasks or even in the medical field, a **False Negative** can be fatal, like telling a sick person they are healthy.